TAKSHASHILA UNIVERSITY

FACULTY OF SCIENCES
SCHOOL OF COMPUTER SCIENCES

Curriculum and Syllabi

Regulation – R25

Academic Year:

2025 - 2026

MASTER OF COMPUTER APPLICATIONS
with specialization in Data Science

II Semester

CURRICULUM

School: School of Computer Science
Programme: Master of Computer Science with specialization in Data Science

Batch: 2025 -2027
Regulation: R-2025

SEMESTER – II										
Sl. No	**Course Code**	**Course Title**	**Category**	**Periods**			**Credits**	**Max. Marks**		
				L	**T**	**P**		**CAM**	**ESM**	**Total**
Theory and Practical (Integrated)										
1	P25CA09B	Advanced Java	DSC	3	0	2	4	40 20	60 30	150
2	P25CA10B	Linux Programming	DSC	3	0	2	4	40 20	60 30	150
3	P25CA82B	Probability and Statistics for Data Science and Data Preparation	DSC	3	0	2	4	40 20	60 30	150
4	P25CM01T	Accounting and Financial Management	MDC	4	0	0	4	40	60	100
5	P25CA11T	Agile Software Development and Testing	DSE	4	0	0	4	40	60	100
6	P25CA14T	Image Processing and Pattern Recognition	DSE	4	0	0	4	40	60	100
Skill Enhancement Course										
7	P25CA17T	Multimedia Tools	SEC	2	0	0	2	100	-	100
						TOTAL	26	400	450	850

BoS Chairman **Dean-FSc** **Dean Academic Affairs**

SEMESTER – II										
Sl. No	**Course Code**	**Course Title**	**Cate gory**	**Periods**			**Cre dits**	**Max. Marks**		
				L	**T**	**P**		**CAM**	**ESM**	**Total**
Discipline Specific Elective - III										
1	P25CA11T	Agile Software Development and Testing	DSE	4	0	0	4	40	60	100
2	P25CA12T	Object Oriented Analysis and Design	DSE	4	0	0	4	40	60	100
3	P25CA13T	Smart Application Development	DSE	4	0	0	4	40	60	100
Discipline Specific Elective - IV										
1	P25CA14T	Image Processing and Pattern Recognition	DSE	4	0	0	4	40	60	100
2	P25CA15T	Theory of Computation	DSE	4	0	0	4	40	60	100
3	P25CA16T	Digital Banking	DSE	4	0	0	4	40	60	100

P25CA09B	Advanced Java	L	T	P	C	Hrs
		3	0	2	4	75

Course Objectives:

- To provide hands-on experience in developing Java applications using RMI.
- To understand GUI programming in Java, including Swing architecture and common GUI components.
- To introduce most of the basic in Servlet Programming.
- To provide concepts and features of JavaServer Pages technology for building dynamic web applications.
- To provide fundamentals of Enterprise JavaBeans (EJB) and their role in enterprise application development.

Course Outcomes:

After completion of the course, the students will be able to

CO1: Able to Understand the RMI architecture and working with RMI

CO2: Understand the GUI components an JDBC with my java.sql

CO3: Understanding Java Servlet and Servlet configuration.

CO4: Understanding concepts of JSP an JSTL

CO5: Able to understand the EJB Architecture and JavaFX technologies.

UNIT I **9 Hours**

Introducing – RMI - Understanding RMI Architecture-Working with RMI-Application Development with RMI-Created Distributed Application Development with RMI.-Remote Object Activation Object Activation. RMI Security using SSL Bypassing.

UNIT II **9 Hours**

GUI Components: Simple GUI Based I/O with JOptionPane - Overview of Swing Components - Displaying Text and Images in a Window - Common GUI Event Types and Listener Interfaces - How Event Handling Works - JButton - JCheckBox - JRadioButton - JList - Multiple Selection Lists - Mouse Event Handling - JPanel Subclass for Drawing with the Mouse - Key -Event Handling- Layout Managers. JDBC and Database Programming: Introduction to JDBC-JDBC Drivers-Using Data Source Object to Make a Connection-JDBC Process with java.sql-The Result Set-JDBC Processes with javax.sql.

UNIT III **9 Hours**

Understanding Servlet Programming: Overview-Features of Java Servlets- Package javax.servlet Description-Servlet Configuration- Servlet Life Cycle-Understanding Response and Request-Reading Form Data from Servlet

UNIT IV **9 Hours**

Understanding of JSP and JSTL: Section A: Understanding Javaserver Pages- Introducing JSP Technology-Understanding Page Life-Cycle-JSP Documents-JSP Elements- JSP Tag Extensions – Tag Libraries.

UNIT V **9 Hours**

Understanding EJB: EJB Fundamentals – EJB Architecture-The EJB Interfaces-EJB Roles-Session Bean-Statefull versus Stateless Session Bean-Developing Session Bean. Entity Bean. Bean Managed Persistence in Entity Beans. Container Managed Persistence-Deployment Descriptor Case Study & Self Study Component: JavaFX Technologies

Total Lecture Hours: 45

Practical Exercises:

1. Develop a complete data manipulation application using Swing/JFC and JDBC.
2. Develop a simple Notepad application using Swing/JFC.
3. Develop an Interest Calculator using Swing/JFC.
4. Develop an application for User Authentication and Personalization using Servlet and JDBC.

5. Develop a simple Shopping Cart using JSP and JDBC.
6. Develop a simple on-line banking using JSP.
7. Develop a complete data manipulation application using JSTL and JDBC.
8. Develop an on-line examination application using JSTL and JDBC.
9. Create Java Beans: GUI Component, Converting GUI Component into a Bean
10. Write a java program to adding an Event set to a Bean

Total Practical Hours: 30

Text Books:

1. Kogent Learning Solutions Inc. "Java Server Programming (J2EE 1.4)" Black Book 2007 Platinum Edition.
2. Paul J. Deitel and Harvey M. Deitel, "Java for Programmers", Pearson Education, 2010 Edition.

Reference Books:

1. James McGovern, Rahim Adatia, Yakov Fain, Jason Gordon, Ethan Henry, Walter Hurst, Ashish Jain, Mark Little, Vaidyanathan Nagarajan, Harshad Oak, Lee Anne Phillips, "Java 2 Enterprise Edition 1.4 Bible", Wiley Publication, 2003 Edition.
2. Herbert Schildt, "The Complete Reference J2EE", Tata McGraw Hill, 2008 Edition.

Web References:

1. https://www.codecademy.com/learn/learn-advanced-java
2. https://forums.oracle.com/ords/apexds/post/jsp-servlets-and-rmi-8551
3. https://www.geeksforgeeks.org/advance-java/jstl-jsp-standard-tag-library/

P25CA10B	Linux Programming	L	T	P	C	Hrs.
		3	0	2	4	75

Course Objectives:

- To understand the Linux programming environment, file system structure, and basic command-line operations.
- To understand the importance of Linux shell programming.
- To understand file and directory maintenance.
- To understand the starting process and process structure.
- To introduce advanced machine learning topics such as neural networks, deep learning, and ensemble methods

Course Outcomes:

After completion of the course, the students will be able to

CO1: Able to understand the Linux file system.

CO2: Understand the Linux shell programming & system programming.

CO3: Understanding the Linux File Structures

CO4: Understanding concepts of process and signals

CO5: Able to understand the Inter-process Communication

UNIT I **9 Hours**

Getting Started: An Introduction to UNIX, Linux, and GNU-Programming Linux. Shell Programming: Features – Definition - Pipes and Redirection.

UNIT II **9 Hours**

Shell programming: The Shell as a Programming Language – Shell Syntax – Going Graphical: The dialog Utility – Putting it all together.

UNIT III **9 Hours**

Working with Files: Linux File Structures–System calls and device drivers–Library functions–Low level file access– file and directory maintenance–scanning directories – Errors

UNIT IV **9 Hours**

Process and signals: Definition – Process Structure – Starting New Processes – Signals.

UNIT V **9 Hours**

Inter- Process Communication: Pipes – Process Pipes – The Pipe Call – Parent and Child Processes. Semaphores, Shared Memory, and Message Queues: Semaphores - Shared Memory - Message Queues

Practical Exercises: **Total Practical Hours: 30**

1. Write a Shell Script with menu driven to check if the input string or the given number is a palindrome or not.
2. Write a Shell Script which displays:
    i. List of all files in the current directory to which you have read, write and execute permissions.
    ii. Receive any number of file names as arguments and whether the argument supplied is a file or directory. If it is a directory it should appropriately reported. If it is a file name, then name of the file as well as the number of lines present in it should be reported.

3. Write a Shell Script to accept a number in the command line and display the sum of the digits of that number and the sum up to that number
4. Write a Shell Script with Menu Driven for computing factorial value of a given number and generating Fibonacci series of the given number of terms using recursive functions
5. Write a Shell Script to prepare pay slip for an employee whose employee number, name, basic pay and loan amount are given as input
6. Create a file called test.dat which contains sample data as follows:

a00001	Shanthi	80
a00007	Arun	70
s00005	Karthi	50

Answer the following questions based on the above data:

i. Display the contents of the file sorted according to the marks in descending order.
ii. Display the names of the students in alphabetical order ignoring the cases.
iii. Display the list of students who have scored marks between 60 and 80.
iv. Display the list of students and their register numbers.

7. Write a Shell Script with Menu Driven for File manipulation which includes
   1) Creating a file 2) Editing a file 3) Removing a file/directory 4) Copying a file 5) Appending contents of files 6) Displaying content of a file and 7) Translating contents of a file either lowercase or uppercase
8. Write a Shell Script to sort the given N numbers and print the biggest and smallest numbers and their corresponding positions.
9. Write a C program for Implementing LINUX Command: pwd
10. Write a C program for Implementing LINUX Command: ls –l
11. Write a C program for Implementing LINUX Command: cp<source_file><destination_file>
12. Write a C program for Implementing LINUX Command: rm<filename | filepattern>
13. Write a C program for Implementing LINUX Command: cat <filename1> <filename2> <filename3>
14. Write an application for producer-consumer using Semaphores.
15. Write an application for client-server model using Semaphores.

Text Books:

1. Neil Matthew, Richard Stones, "Beginning Linux Programming", 4th Edition, WROX, 2011.

Reference Books:

1. Uresh Vahalia, "UNIX Internals, The New Frontiers", Pearson Education Limited, 2002.
2. Peter Dyson, Stan Kelly–Bootle, John Heilborn, "UNIX Complete", BPB Publications, 1999.

Web References:

1. https://www.codecademy.com/learn/learn-advanced-java

P25CA82B	Probability and Statistics for Data Science	L	T	P	C	Hrs
		3	0	2	4	75

Course Objectives: The objective of this course is to

- To introduce fundamental concepts of statistics, probability, and data measurement for data analysis.
- To develop the ability to collect, organize, and summarize data using statistical techniques.
- To understand probability distributions and random variables for modeling real-world problems.
- To apply descriptive and inferential statistical methods for data interpretation.
- To perform exploratory data analysis and hypothesis testing using Python tools.

Course Outcomes: Through this course student should be able to:
CO1: Explain fundamental concepts of statistics and probability, including data types, measurement scales, and probability laws. Students will be able to relate these concepts to real-world business and data science problems.
CO2: Apply descriptive statistical techniques such as measures of central tendency, dispersion, skewness, and kurtosis. Students will also be able to represent data effectively using appropriate charts and graphical methods.
CO3: Analyse random variables and probability distributions, including discrete and continuous distributions. Students will be able to compute expectation, variance, covariance, and interpret distribution functions for data modeling.
CO4: Students will be able to apply discrete and continuous probability distributions and explain the Central Limit Theorem for analyzing real-world data problems.
CO5: Perform statistical inference using hypothesis testing techniques such as Z-test, t-test, F-test, and Chi-square test. Conduct exploratory data analysis using Python libraries such as NumPy, Pandas, Matplotlib, and Seaborn. Students will be able to perform hypothesis testing, interpret patterns, relationships, and trends in datasets to support data-driven decision-making.

UNIT I **9 Hours**
Introduction to Statistics: Statistics in Business - Basic Statistical Concepts - Data Measurement.
Charts and Graphs: Frequency distributions - Quantitative data graphs - Qualitative data graphs - Graphical representation of 2-variable numerical data
Descriptive Statistics: Measures of central tendency - Measures of Dispersion - Measures of Skewness - Measures of Kurtosis

UNIT II **Theory and Types of Probability** **9 Hours**
Introduction to Probability - Methods of assigning probabilities - Structure of probability
Marginal, Union, Joint and conditional probabilities - Addition laws - Multiplication laws - Bayes Rule

UNIT III **Random Variables and Distribution functions** **9 Hours**
Random Variables - Discrete and Continuous; Distribution functions (PDF, CDF and PMF) and its properties. Bivariate Random Variables - Joint Probability functions, marginal distributions, conditional distribution functions, Independence of Random variables. Expectation, Variance, and Co-variance of random variables.

UNIT IV **9 Hours**
Discrete Distributions : Discrete vs Continuous distributions - Binomial distribution - Poisson distribution - Hypergeometric distribution**. Continuous Distributions:** The uniform distribution - Normal distribution - Approximating binomial distribution using the normal curve - Exponential distribution. **Central Limit Theorem (CLT)**: Sampling distribution of mean and its importance

UNIT V **Exploratory Data Analysis and Hypothesis Testing** **9 Hours**
Concept of statistical hypothesis, Hypothesis testing, Tests based on Z, t, F, Chi-square test, goodness of fit, test of independence, Plotting graphs using Pandas, Numpy for EDA, Data Visualization using Matplotlib and

seaborn - Basic Charts and Diagrams, Heatmaps: Visualizing Correlations, Multidimensional visualization: Adding variables, Manipulations, Reference, Multivariate plots - Real-world Data Exploration and Statistics Case Study

Practical Exercises: **Total Practical Hours: 30**

1. Descriptive Statistics Using NumPy and Pandas
2. Data Type Identification and Frequency Distribution Analysis
3. Probability Fundamentals Using Python Simulation
4. Random Sampling and Sample vs Population Analysis
5. Normal Distribution and Z-Score Analysis
6. Binomial Distribution Implementation and Analysis
7. Poisson Distribution and Event Modeling
8. Correlation Analysis Between Numerical Variables
9. Covariance Analysis and Interpretation
10. Hypothesis Testing Using Z-Test and T-Test
11. Confidence Interval Estimation for Population Mean
12. Univariate Exploratory Data Analysis
13. Bivariate and Multivariate Exploratory Data Analysis
14. Complete Exploratory Data Analysis Case Study

Text Books:

1. Practical Statistics for Data Scientists. “Peter Bruce, Andrew Bruce, Peter Gedeck”.
2. Think Stats. “B. Downey”
3. Exploratory Data Analysis with Python. “Nunes & de Castro”

Reference Books:

1. Naked Statistics: Stripping the Dread from the Data Book by Charles Wheelan
2. Storytelling with Data. “Cole Nussbaumer Knaflic”

P25CM01T	**Accounting and Financial Management**	L	T	P	C	Hrs
		4	0	0	4	60

Course Objectives

- To understand the concept of financial Accounting
- To gain knowledge about preparation of final accounts.
- To Analysis the ratio types and Return on Investments in financial decision
- To know financial management and EBIT analysis process
- To Deal with the types of Capital Budgeting

Course Outcomes

After completion of the course, the students will be able to

CO1 - Explain the fundamentals of Accounting

CO2 - Preparation of final accounts

CO3 - Analyze the Ration Analysis

CO4 - Identity financial decision and management concepts

CO5 - Analysing the capital budgeting and measures of budgeting concepts

UNIT – I Introduction 12 Hours

Definition of an accounting, characteristics of accounting, objectives of keeping books of accounting, Advantages of book keeping, double entry system of accounting, concepts and convention - Journal - ledger - Preparation of trial balance.

UNIT - II Final Accounts 12 Hours

Preparation of final accounts (sole proprietary, firm concern only), trading accounts, profit and loss Accounts and balance sheet.

UNIT - III Ratio Analysis 12 Hours

Ration Analysis: Definition, significance of ratio analysis, types of ratios-return on investments, profitability ratio, Turnover ratio, limitations of ratios.

UNIT - IV Financial Management 12 Hours

Financial management- Meaning and role of financial management- dividend decision – investment decision – financial decision – capital structure – EBIT analysis – Leverages.

UNIT - V Capital Budgeting 12 Hours

Capital Budgeting – method and measures analysis - payback period method – average rate of return – Net present value methods – internal rate of return method.

Text Book:

1. S.P. Jain & K.L. Narang, "Financial Accounting", Kalyani Publishers, 12^{th} Edition, 2014.
2. P.C. Tulsian & Bharat Tulsian, "Financial Accounting", S. Chand, 2^{nd} Edition, 2016.

Reference Books:

1. Elements of book keeping & Accountancy by B. S. Shah & Sons.
2. Financial Accounting & Management (For BCA) – by B. S. Shah Prakahsan.
3. Financial Management – by I.M. Pandey, Vikas Publication.
4. Financial Management by Jonathan Berk, Peter DeMarzo and Rama Seth, Pearson Education.

P25CA11T	Agile Software Development and Testing	L	T	P	C	Hrs
		4	0	0	4	60

Course Objectives:

- To know about the Software Engineering and process
- To study about the Software Testing & Strategies
- To understand about the Agile software development Process
- To get knowledge in Agile methodology and
- To get adequate knowledge in Agile global development & Quality Assurance

Course Outcomes:

After completion of the course, the students will be able to

CO1: Understand about the Software Engineering and metrics.
CO2: Understand about the Software testing and strategies
CO3: Identify the Agile process and driven development
CO4: Develop technique and methodology to improve the software development.
CO5: Show how the agile process can be used in the quality assurance.

UNIT I: Introduction to Software Engineering **12 Hours**

Introduction to Software Engineering: Overview of Software Engineering- Software Process – Software Product – Software Process Models – Software Project Planning – Project Scheduling and Tracking – Software Metrics - System Engineering – Business Process Engineering – Product Engineering – Requirement Engineering – Software analysis and Principles – Software architecture – Design concepts and principles – User interface design

UNIT II: Software Testing **12 Hours**

Software Testing: Introduction to Software Testing – Testing Principles – Need for Testing – Testing strategies - Software Testing techniques – Psychology of testing – White box testing – Black Box testing – Gray Box testing– Robustness testing - Verification and Validation testing – Syntax testing – Unit testing – Integration testing – System testing – levels of testing.

UNIT III: Introduction to Agile Process **12 Hours**

Introduction to Agile Process: - Introduction Agile - Concepts of Agile process - Agile Driven development - Scrum: Concepts – deliverable and methods. Extreme Programming: Concepts – deliverable – methods. Unified Process: Concepts – deliverable - methods. EVE: Concepts – methods – deliverable.

UNIT IV: Agile Technology and Methodology 12 **Hours**

Agile Technology and Methodology: Concepts of Agile management – Agile Principles – Agile Software model –Agile Method classification -Agile Project management –- Agile Ethics – Agile in Design, Testing, Documentation – Agile Drivers.

UNIT V: Knowledge Management and Quality Assurance 12 **Hours**

Agile Knowledge Management and Quality Assurance: Introduction to Agile Information systems – Agile development – Knowledge Management in Software Engineering – Issues and challenges of Agile methodology – Agile product development – Agile metrics – Agile Quality Assurance – Agile global software development

Text Books:

1. Roger S. Pressman, Software Engineering. A Practitioners Approach, Fifth Edition, 2001
2. Richard Farley, Software Engineering Concepts, Tata McGraw Hill, 2003.
3. William E.Perry, *" Effective Methods for Software Testing (2nd Edition) "*,John Wiley & Sons, 2000.
4. Craig Larman, "Agile and Iterative Development A Manger's Guide" Pearson Education,First Edition, India, 2004.
5. DavidJ.Anderson and Eli Schragenheim," Agile Manament for Software Engineering: applying the theory of constraints for business results",Prentice Hall, 2003.

Reference Books:

1. GlenfordJ.Myers, *"The Art of Software Testing "*, John Wiley & Sons, 1997.
2. Boris Beizer, Black-Box Testing: *"Techniques for Functional Testing of Software and Systems"*,John Wiley & Sons, 1995.
3. P.C.Jorgensen, *" Software Testing - A Craftman's Approach ",* CRC Press, 1995.
4. C. Ghezzi, M. Jazayeri and D. Mandrioli, Fundamentals of Software Engineering, Printice Hall of India Private Limited, 2nd Edition,2002.
5. Shore, "Art of Agile Development", Shroff Publishers & Distributors, 2007.
6. KevinC.Desouza "Agile Information systems:" conceptualization, construction and Management", Butterworth- Heinemann, 2007

Web References:

1. https://udemy.com
2. https://udacity.com
3. https://www.digimat.in/
4. https://nptel.ac.in/courses/

P25CA12T	Object Oriented Analysis and Design	L	T	P	C	Hrs
		4	0	0	4	60

Course Objectives:

- To make aware of the software requirements, design the software using tools
- To be acquainted with the writing of test cases using different testing techniques.
- To get familiarized to the usage of UML tool kit.
- To understand the requirements of the software and to map them appropriately to subsequent phases of the software development
- To develop the ability to verify and validate their designs

Course Outcomes:

After completion of the course, the students will be able to

CO1: The students should be able to specify software requirements, design the software using tools

CO2: To write test cases using different testing techniques.

CO3: Students must be able to analyses and design the problem at hand.

CO4: Students should be able to use UML tools for the designing the software and test the correctness and soundness of their software through testing tools.

CO5: Students should be able to develop, verify and validate their designs

UNIT I **12 Hours**

Object Orientation – System development – Review of objects - inheritance – Object relationship – Dynamic binding – OOSD life cycle – Process – Analysis – Design – prototyping – Implementation – Testing- Overview of Methodologies.

UNIT II **12 Hours**

Rambaugh methodology, OMT – Booch methodology, Jacobson methodology – patterns – Unified approach – UML – Class diagram – Dynamic modelling.

UNIT III **12 Hours**

Introduction - UML – Meta model - Analysis and design - more information. Outline Development Process: Overview of the process - Inception - Elaboration-construction - refactoring patterns transmission - iterative development - use cases.

UNIT IV **12 Hours**

OO Design axioms – Class visibility – refining attributes – Methods –Access layer – OODBMS – Table – class mapping view layer.

UNIT V **12 Hours**

Interaction diagram-package diagram-state diagram-activity diagram-deployment diagram - UML and programming.

Text Books:

1. Ali Bahrami, “Object Oriented System Development”, McGraw-Hill International Edition 2017.
2. Martin Fowler, Kendall Scott, "UML Distilled", Addision Wesley
3. Eriksson, "UML Tool Kit", Addison Wesley

Reference Books:

3. James McGovern, Rahim Adatia, Yakov Fain, Jason Gordon, Ethan Henry, Walter Hurst, Ashish Jain, Mark Little, Vaidyanathan Nagarajan, Harshad Oak, Lee Anne Phillips, “Java 2 Enterprise Edition 1.4 Bible”, Wiley Publication, 2003 Edition.
2. Herbert Schildt, “The Complete Reference J2EE”, Tata McGraw Hill, 2008 Edition.

P25CA13T	Smart Application Development	L	T	P	C	Hrs
		4	0	0	4	60

Course Objectives:

After completing this course, students will be able to:
CO1: Understand the fundamentals of mobile computing, evolution of smart applications, and the architecture of the Android platform including SDK setup and development lifecycle.
CO2: Analyze the structure of Android applications, including activities, services, intents, Android Manifest file, permissions, and intent filters for building functional applications.
CO3: Design and develop effective user interfaces using layouts, screen elements, drawing techniques, and animations to enhance user experience.
CO4: Apply testing techniques and deployment strategies, manage application preferences and resources efficiently, and prepare applications for publishing.
CO5: Utilize Android APIs for data storage (SQLite), content providers, networking, web services, telephony services, and deploy Android applications globally.

Course Outcomes:

- Understand the Android development environment.
- Identify the basic concepts in App development.
- Critically evaluate mobile applications on their design aspects
- Develop a simple mobile application.
- Design a user interface using the Android framework.
- Implement advanced features in Android applications.
- Understand the principles of testing and deploying Android applications.
- Familiarize with advanced features and various applications in Android development.

UNIT I **12 Hours**

Introduction to Android Basics: Overview of smart applications and their role in modern computing, Evolution of mobile computing, Introduction to mobile application frameworks (e.g., Android, iOS). User interface design for mobile applications, Mobile application development lifecycle, Overview of the Android platform and SDK, Eclipse installation and Android setup, Building the first Android application.

UNIT II **12 Hours** *App*

Development Essentials: Anatomy of Android applications, Overview of the Android Manifest file Android terminologies and application context, Components: activities, services, and intents, Android Manifest file settings and permissions, utilizing intent filters and managing permissions.

UNIT III **12 Hours**

UI Design Fundamentals: Introduction to user interface screen elements, designing user interfaces with layouts, drawing techniques and working with animation.

UNIT IV **12 Hours**

Testing and Deployment: Principles of testing Android applications, Strategies for publishing Android applications, Utilizing Android preferences, managing application resources in a hierarchy, Working with different types of resources.

UNIT V **12 Hours**

Common Android APIs: Using Android data and storage APIs, managing data using SQLite, sharing data between applications with content providers, Using Android networking APIs, Using Android web APIs, Using Android telephony APIs, Deploying Android applications to the world.

Text Books:

1. Lauren Darcey and Shane Conder, "Android Wireless Application Development," Pearson Education, 2nd ed. (2011).
2. Reto Meier, "Professional Android 2 Application Development", Wiley India Pvt. Ltd.

Reference Books:

1. Mark L Murphy, "Beginning Android", Wiley India Pvt. Ltd.
2. Barry Burd, "Android Application Development All in One for Dummies".
3. Grant Allen and S.Ann Mc Fadden, "Beginning Android 4 Application Development", Wiley.

Web Reference:

1. http://index-of.es/Android/Android_Application_Development_For_Dummies.pdf
2. https://developer.android.com/docs

P25CA14T	Image Processing and Pattern Recognition	L	T	P	C	Hrs
		4	0	0	4	60

Course Objectives:

- Understand the fundamental concepts of digital image processing and mathematical tools for image analysis.
- Learn essential image enhancement, segmentation, and representation techniques.
- Comprehend feature extraction, dimensionality reduction, and classification methods for pattern recognition.
- Explore supervised and unsupervised learning techniques for pattern classification.
- Analyze real-world applications of image processing and pattern recognition.

Course Outcomes:

After completion of the course, the students will be able to

CO1: Demonstrate a deep understanding of image processing techniques in both spatial and frequency domains

CO2: Develop skills to apply image enhancement and restoration techniques.

CO3: Analyze and implement segmentation, feature extraction, and representation methods.

CO4: Understand and apply statistical and machine learning approaches to pattern recognition.

CO5: Solve real-world problems using modern image processing and pattern recognition techniques.

UNIT I **12 Hours**

Fundamentals of Digital Image Processing: Introduction to Digital Image Processing: Historical development and applications of DIP. Components of an image processing system. Digital Image Fundamentals: Visual perception principles and image formation models. Sampling, quantization, and image resolution. Pixel relationships (neighborhoods, connectivity, adjacency). Mathematical Preliminaries: Linear and nonlinear operations. Basics of convolution and correlation. Tools and Platforms for Image Processing: (MATLAB/OpenCV Basics) .

UNIT II **12 Hours**

Image Enhancement and Restoration: Image Enhancement in Spatial Domain: Point operations (gray-level transformations). Histogram processing and equalization. Spatial filtering techniques (smoothing and sharpening). Image Enhancement in Frequency Domain: Fourier transform and discrete Fourier transform (DFT). Frequency domain filters (low-pass, high-pass, bandpass). Image Restoration: Image degradation/restoration process. Noise models (Gaussian, impulse, etc.). Wiener and inverse filtering techniques.

UNIT III **12 Hours**

Image Segmentation and Feature Extraction: Image Segmentation Techniques: Edge detection (Sobel, Canny). Thresholding (global, adaptive, Otsu's method). Region-based segmentation (region growing, region splitting/merging). Feature Extraction and Dimensionality Reduction: Corner detection (Harris), texture features (GLCM). SIFT, SURF, HOG, and Gabor filters. PCA and LDA for feature reduction. Image Representation: Boundary descriptors (perimeter, compactness). Regional descriptors (moments, textures).

UNIT IV **12 Hours**

Pattern Recognition Basics: Introduction to Pattern Recognition: Components of a pattern recognition system. Supervised vs. unsupervised learning. Feature space, feature selection, and dimensionality. Statistical Pattern Recognition: Parametric classification (Bayes classifier). Non-parametric techniques (k-NN, Parzen window).

Unsupervised Learning Methods: Clustering techniques (k-means, hierarchical clustering). Evaluation metrics for clustering.

UNIT V **12 Hours**

Advanced Pattern Recognition and Applications: Fuzzy and Syntactic Pattern Recognition: Fuzzy logic in pattern classification. Fuzzy clustering (FCM). Grammar-based syntactic pattern recognition. Performance Evaluation: Risk, error probabilities, and ROC curves. Accuracy, precision, recall, F1 score. Applications of Pattern Recognition: Face recognition, object detection, and character recognition. Medical imaging, remote sensing, and biometrics

Text Books:

1. Rafael C. Gonzalez and Richard E. Woods: "Digital Image Processing", 4th Edition, Pearson, 2018.
2. Sergios Theodoridis, Konstantinos Koutroumbas: "Pattern Recognition", 5th Edition, Academic Press, 2018.

Reference Books:

1. Milan Sonka, Vaclav Hlavac, Roger Boyle: "Image Processing, Analysis, and Machine Vision", 4th Edition, Cengage Learning, 2014.
2. Christopher M. Bishop: "Pattern Recognition and Machine Learning", Springer, 2016.
3. Richard O. Duda, Peter E. Hart, David G. Stork: "Pattern Classification", 2nd Edition, Wiley, 20
4. Scott E. Umbaugh: "Digital Image Processing and Analysis", 3rd Edition, CRC Press, 2017

Web References:

1. https://ieeexplore.ieee.org/document/1674602/
2. https://pyimagesearch.com/
3. https://docs.opencv.org/4.x/d9/df8/tutorial_root.html
4. https://refhub.ir/refrence_detail/image-processing-and-pattern-recognition-fundamentals-and-techniques/

P25CA15T	Theory of Computation	L	T	P	C	Hrs
		4	0	0	4	60

Course Objectives:

CO1: Understand the basic concepts of Automata Theory, including alphabets, strings, languages, and grammars, and analyze finite state machines such as DFA, NFA, Moore, and Mealy machines.
CO2: Apply regular expressions and finite automata techniques for language representation and conversion between automata and regular expressions using algebraic laws and Arden's Theorem.
CO3: Analyze Context-Free Grammars (CFG), construct parse trees, remove ambiguity, convert grammars into normal forms (CNF & GNF), and understand Pushdown Automata (PDA) and their equivalence with CFG.
CO4: Understand the design and working of Turing Machines, explore different types of TMs, and explain recursive and recursively enumerable languages including the halting problem.
CO5: Distinguish between decidable and undecidable problems, understand Linear Bounded Automata (LBA), Context-Sensitive Languages (CSL), and explain the Chomsky Hierarchy and relationships among language classes.

Course Outcomes:

After completion of the course, the students will be able to

- Develop a formal notation for strings, languages and machines and to design finite automata to accept a set of strings of a language.
- Analyze and generate regular expressions for any structure and demonstrate that a given language is regular or not.
- Design context free grammars to generate strings from a context free language and create Push Down Automata for any context free grammar.
- To prove equivalence of languages accepted by Linear Bound Automata and languages generated by Context sensitive grammars.
- To study the concept of Turing machines and analyze how various problems are solved using digital computers.

UNIT I **12 Hours**

Concepts of Automata Theory: Strings, Alphabet, Language, Grammar - Finite state systems: NFA, DFA, Definitions, Equivalence of NFA and DFA, NFA to DFA conversion, NFA with epsilon transitions, Minimization of DFA, Designing Moore and Mealy machines.

UNIT II **12 Hours**

Regular expressions, Finite Automata & Regular Expression operations, Conversion of Finite Automata to Regular expressions. Converting Regular Expressions to Automata - Arden's Theorem –Algebraic Laws for Regular Expressions. Pumping Lemma for Regular Languages, Application of Regular Expressions.

UNIT III **12 Hours**

Context free grammar, Definition, Derivation. Parse Trees and Ambiguity in Grammars - Conversion to Normal Forms: Chomsky normal form, Greibach normal form - Pumping Lemma for Context Free Languages - Properties of Context Free Language - Pushdown Automata: Definition, Acceptance of PDA, Deterministic PDA, Equivalence of PDA and CFG.

UNIT IV **12 Hours**

Turing Machines: Definition, Transition diagram, Design & Roles of Turing machine, Church - Turing Thesis, Modular Construction of complex Turing machines, Types of Turing machines, Extensions of Turing machines, Universal Turing machines, Recursive and recursively enumerable languages, halting problem of TM.

UNIT V **12 Hours**

Decidable and Undecidable problems. Introduction to Linear Bounded Automata Definition – Context-sensitive languages (CSL) - Context-sensitive grammars (CSG), LBA and equivalence with CSG. Chomsky Hierarchy, Relation between languages.

Text Books:

1. Hopcroft J. E., Motwani,R. and Ullman J.D., Introduction to Automata Theory, Languages, and Computation, 3rd Edition, ISBN: 978-03-214-5536-9.
2. PadmaReddy,A.M., Finite Automata and Formal Languages, 1st Edition, Pearson Education, ISBN: 978-81-317-6047-5.

Reference Books:

1. Mishra,K.L.P. and Chandrasekaran,N, Theory of Computer Science, Automata, Languages and Computation, 3rd Edition, PHI, 2014, ISBN: 978-81-203-2968-3.
2. Peter Linz, An Introduction to Formal Languages and Automata, 4th Edition, Narosa Publishing Co., ISBN: 978-81-7319-781-9
3. John.C.Martin, Introduction to Languages and the theory of computation, 3rd Edition, Tata Mc Graw Hill, ISBN: 978-0-07-066048-9.

Web Reference:

1. https://ocw.mit.edu/courses/18-404j-theory-of-computation-fall-2020/pages/lecture-notes/

P25CA16T	Digital Banking	L	T	P	C	Hrs
		4	0	0	4	60

Course Objectives:

- To acquaint students with knowledge of Digital Banking Products.
- Enable the students to understand the knowledge of Digital Payment System
- Impart the students to understand the new concepts of Mobile and Internet Banking
- Enables the students to have depth knowledge in point of sale terminals
- To understand the ATM and cash deposit system

Course Outcomes:

After the successful completion of the course, the students will be able to:

CO1: Explain the need for digital banking products and the usage of cards.

CO2: Classify the usage of various payment systems.

CO3: Discuss the profitability, risk management and frauds of mobile and internet banking.

CO4: Analyse the approval processes of POS terminals.

CO5: Explain the product features and services of ATM and Cash Deposit Machine.

UNIT I Digital Banking Products 12 Hours

Digital Banking – Meaning – Features - Digital Banking Products - Features - Benefits – Bank Cards –Features and Incentives of Bank cards - Types of Bank Cards - New Technologies - Europay, Master and Visa Card (EMV) - Tap and Go, Near Field Communication (NFC) etc. - Approval Processes for Bank Cards – Customer Education for Digital Banking Products - Digital Lending – Digital Lending Process - Non-Performing-Asset (NPA).

UNIT II Payment System 12 Hours

Overview of Domestic and Global Payment systems - RuPay and RuPay Secure - Immediate Payment Service (IMPS) – National Unified USSD Platform (NUUP) - National Automated Clearing House (NACH) - Aadhaar Enabled Payment System (AEPS) – Cheque Truncation System (CTS) – Real Time Gross Settlement Systems (RTGS) – National Electronic Fund Transfer(NEFT) - Innovative Banking & Payment Systems.

UNIT III Mobile and Internet Banking 12 Hours

Mobile & Internet Banking - Overview – Product Features and Diversity - Corporate and Individual Internet Banking Integration with e-Commerce Merchant sites, IMPS - Profitability - Risk Management and Frauds - Cyber Crime - Cyber Security - Blockchain Technology - Types - Crypto currency and Bitcoins

UNIT IV Point of Sale Terminals 12 Hours

Point of Sale (POS) Terminals - Overview - Features - Approval processes for POS Terminals - Key Components of POS - Hardware - Software - User Interface Design - Cloud based Point of Sale – Cloud Computing - Benefits of POS in Retail Business.

UNIT V Automated Teller Machine and Cash Deposit Systems 12 Hours

Automated Teller Machine(ATM) - Cash Deposit Machine(CDM)& Cash Recyclers - Overview - Features - ATM Instant Money Transfer Systems - National Financial Switch (NFS) -Various Value Added Services - Proprietary, Brown Label and White Label ATMs - ATM & CDM Network Planning - Onsite / Offsite - ATM security, Surveillance and Fraud Prevention.

Text Books:

1. IIBF, 2019.Digital Banking. Taxmann Publications, New Delhi
2. Gordon E. &Natarajan S. 2017 Banking Theory, Law and Practice. 24th Revised Edition. Himalaya Publishing House, New Delhi
3. Ravindra Kumar and Manish Deshpande. 2016 E-Banking. Pacific Books International, 2016.
4. Uppal R.K. 2017 E-Banking: The Indian Experience. Bharti Publications, 2017.

Reference Books:

1. Arunajatesan S, 2017, Technology in Banking, Margham Publications, Chennai..
2. Digital Banking, 2016, Indian Institute of Banking and Finance, Pvt. Limited, New Delhi.
3. Indian Institute of Banking and Finance, 2016 ,General Bank Management, McMillan, Mumbai.
4. SubbaRao.S and Khanna.P.L, 2014, Principles and Practice of Bank Management, Himalaya Publishing House, Mumbai.

Web References:

1 https://ebooks.lpude.in/commerce/bcom/term_4/DCOM208_BANKING_THEORY_AND_PRACTICE.pdf
2 http://www.himpub.com/documents/Chapter1859.pdf.

P25CA17T	Multimedia Tools	L	T	P	C	Hrs
		2	0	0	2	30

Course Objectives:

- Understand the fundamental concepts of multimedia tools for image analysis.
- Learn essential image enhancement and color modelling
- Learn concepts video and audio signals
- Explore compression of multimedia files
- Analyze the techniques of video compression

Course Outcomes:

After completion of the course, the students will be able to

CO1: Understand on the concepts related to Multimedia.
CO2: Understand the concepts like image data representation and color modes.
CO3: Understand the different types of video signals and digital audio.
CO4: Understand multimedia data compression types and audio compression standards.
CO5: Solve basic video compression techniques.

UNIT I : Introduction to Multimedia **6 Hours**
Multimedia - Components of Multimedia System - Multimedia and Hypermedia - Multimedia Authoring – metaphors - Multimedia Production - Multimedia Presentation - Automatic Authoring

UNIT II: Image Data Representations and Color Model **6 Hours**
Color science Human Vision Image data types: Black & white images - 1-bit images (Binary image) - 8-bit (Gray -level images) - Color images - 24-bit color images - 8-bit color images - Color models

UNIT III: Fundamental Concepts in Video **6 Hours**
Types of Video Signals - Analog Video - Digital Video - Basics of Digital Audio: Sound - Digitization of Sound - Quantization and Transmission of Audio - Pulse Code Modulation - Differential coding of Audio - Predictive coding

UNIT IV: Multimedia Data Compression **6 Hours**
Introduction- Basics of Information Theory - Lossless Compression Algorithms - Fix-Length Coding - Run-length coding - Dictionary-based coding - Variable Length Coding - Huffman Coding Algorithm - Audio Compression standards: Introduction - Psychoacoustics model - MPEG Audio

UNIT V: Basic Video Compression Techniques **6 Hours**
Introduction to Video Compression - Video compression standard H.261 - Video compression standard MPEG-1

Text Books:

1. Fundamentals of Multimedia by Ze-Nian Li & Mark S. Drew. Publisher: Prentice Hall

Reference Books:

1. An introduction to digital multimedia by Savage, T. M. and Vogel, K. E. 2008.
2. Digital Multimedia by Nigel Chapman & Jenny Chapman. 2009.

Web References:

1. https://ksuit342.wordpress.com/lectuers/
2. https://www.tutorialspoint.com/multimedia